# YUXUAN HOU

Shenzhen, China | 12413104@mail.sustech.edu.cn | tsawke.github.io



## Education

### Southern University of Science and Technology (SUSTech) — Expected Jun 2028

*B.Eng. in Computer Science and Technology — Turing Class (Honors Program)* — *Shenzhen, China*

**Academic-year GPA (2025–2026): 3.86/4.00** | **Class President**, 2024 CSE Turing Class

**Honors Coursework:** Discrete Mathematics; Data Structures and Algorithm Analysis; Algorithm Design and Analysis; Principles of Database Systems; Artificial Intelligence.

**Other Completed Coursework:** Computer Organization; Computer Vision; Natural Language Processing; Digital Logic; Introduction to Mathematical Logic; Introduction to Computer Programming; Computational Ethics; Group Projects I.

**In Progress:** Operating Systems (Honors); Compiler Principles; Computer Networks; Advanced Computer Programming; Probability and Statistics for Engineering.

## Research Affiliation

**Lab:** **Advanced Intelligent Mobile Systems (AIMS) Lab**, SUSTech

**Advisor:** **Jin Zhang**, Associate Professor, Department of Computer Science and Engineering

## Accepted Publication

**Yuxuan Hou**, Jiao Li, Linshan Jiang, and Jin Zhang. *Oto-Meal: Earable Sensing with PPG and IMU for Personalized Meal Awareness*.
**WellComp 2026 (UbiComp/ISWC 2026 Workshop)** — Computing for Well-being.
Accepted; forthcoming in *UbiComp Companion '26*. DOI: 10.1145/3798063.3841767. arXiv:2608.16678.

## Research Experience

### Oto-Meal: Earable Sensing with PPG and IMU for Personalized Meal Awareness — Sep 2025–Present

*Project Lead & Primary Developer* — *Wearable sensing*

First-author paper accepted to **WellComp 2026 (UbiComp/ISWC 2026 Workshop)**.

**Project Lead** — 2026 National Undergraduate Innovation and Entrepreneurship Training Program (**National Level**; CNY 25,000)

**Project Lead** — 2026 SUSTech "Climbing Program" Project (CNY 15,000; completed)

- Built an audio- and image-free earable prototype using photoplethysmography (PPG) and inertial sensing; evaluated a pooled neural classifier separately from within-user memory matching on **538 recordings and 7,028 windows from 7 participants**.
- Continue follow-up research on data-efficient recognition and transfer learning; completed a **32-case pilot** on cross-user evaluation.

### Neuromorphic Sensing for Edge Devices — Ongoing

*Method Development & Device Evaluation* — *In preparation*

- Develop spiking neural network (SNN) approaches for wireless and wearable sensing, studying accuracy and resource trade-offs across **ten datasets**.
- Evaluate on **Jetson AGX Orin** across three datasets and three seeds, auditing **81 inference conditions** for accuracy, latency, throughput and power measurements.

### LeaseTurn: Tool-Result Scheduling for Language Model Agents — Ongoing

*Systems Research & Prototyping* — *In preparation*

- Investigate tool-result scheduling for language-model agents, focusing on inference latency and resource coordination.
- Build a real-inference prototype and controlled evaluation with **631 paired workflow comparisons**, analyzing scheduling behavior under varied workloads.

## Research Experience (continued)

**3D Acoustic Field Reconstruction and Adaptive Acoustic Modeling** **May 2026–Present**

*Acoustic Modeling & Simulation* *In preparation*

- Study acoustic scene reconstruction from sparse measurements and measurement-geometry calibration; reproduce prior work, develop simulation and evaluation workflows, and assess reconstruction and calibration behavior.

**Related computer vision project:** Team Leader, four-person Object-Level 3D Gaussian Splatting project (Spring 2026); built reconstruction, segmentation and editing pipeline, registering 213/213 frames with 0.71-pixel reprojection error.

**InterveneCompose** (ongoing; manuscript in preparation): theoretical study of multi-label function properties, query complexity and exact computational validation.

**Q-CalTrack** (ongoing; manuscript in preparation): wireless motion sensing, signal-processing evaluation and uncertainty analysis.

**CUMCM Research** (ongoing; manuscript in preparation): data-efficient learning for combinatorial search, with controlled experimental comparisons.

**RydbergTrack** (Spring semester 2026; Team Leader, three-person project): built a reproducible MATLAB pipeline for 150 million samples; separated 3 static and 12 motion recordings within the project dataset.

## Technical Skills

**Programming:** Python, C/C++, MATLAB, Java, Verilog
**ML & Scientific Computing:** PyTorch, scikit-learn, NumPy, SciPy; spiking neural networks, signal processing, numerical optimization
**Systems & Tools:** CUDA, COLMAP, Vivado, Git, Linux, LaTeX; inference scheduling, reproducible evaluation, automated testing

## Selected Honors & Awards

**2025** **National First Prize**, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)
**2025** **Silver Medal (Rank 4)**, International Collegiate Programming Contest (ICPC)
**2025** **Silver Medal**, China Collegiate Programming Contest (CCPC)
**2025** **Gold Medal**, Guangdong Collegiate Programming Contest
**2026** **International First Prize (1st Place)**, Application Development Division Finals, 1st Hong Kong International AIGC Cultural Digital Content Creation Competition
**2025** Provincial First Prize, 16th Lanqiao Cup, Guangdong; C/C++ Programming, University Group A
**2024/25** SUSTech University Scholarship; Zhiren College Motto Scholarship “Rixin”; SUSTech Outstanding Student Leader
**2026** Undergraduate Talent Development Excellence Program Grant (2024 Cohort; CNY 6,000)
**2025/26** SUSTech Outstanding Communist Youth League Cadre
**2026** Second Prize, CCF Service Computing Innovation Competition / Sangfor CoStrict AI-Native Innovation Competition; *GeoAgent* (team project)

**Earlier informatics honors:** Provincial Rank 9, National Olympiad in Informatics (NOI) Joint Provincial Selection; First Prizes in the NOI Spring Test and CCF CSP-S (senior-level certification).

## Leadership & Service

**President, Student Union | Chair, Student Development Committee:** lead organizational planning, team coordination, recruitment, student activities, member development, leadership handovers, and operating procedures.

**Head, Zhiren College Peer Mentor Team:** lead 15 mentors serving 231 incoming students across 5 classes and outreach covering 25 provinces; organize onboarding, orientation, and ongoing support.

**Collective honors:** SUSTech Outstanding Student Union; First Prize, Zhiren College Outstanding Student Organization (2025–2026).

**Volunteer service: 87.17 verified hours**, including campus events and accessibility initiatives.

**Other roles:** Turing Class President; 3D Manufacturing Club Vice President; competitive programming, supercomputing, rowing, emcee, and exhibition hall docent teams.